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Gentamicin (systemic): Drug information

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For additional information see "[Gentamicin \(systemic\): Patient drug information](#)" and "[Gentamicin \(systemic\): Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

ALERT: US Boxed Warning

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Toxicity:



Ensure that patients treated with aminoglycosides are under close clinical observation because of the potential toxicity associated with their use.

As with other aminoglycosides, gentamicin is potentially nephrotoxic. The risk of nephrotoxicity is greater in patients with impaired renal function and in those who receive high dosages or prolonged therapy.

Neurotoxicity, manifested by ototoxicity, both vestibular and auditory, can occur in patients treated with gentamicin, primarily in those with preexisting renal damage and in patients with healthy renal function treated with higher doses and/or for longer periods than recommended. Aminoglycoside-induced ototoxicity is usually irreversible. Other manifestations of neurotoxicity may include numbness, skin tingling, muscle twitching, and convulsions.

Closely monitor renal and eighth cranial nerve functions, especially in patients with known or suspected reduced renal function at onset of therapy, and also in those whose renal function is initially normal but who develop signs of renal dysfunction during therapy. Examine urine for decreased specific gravity, increased excretion of protein, and the presence of cells or casts. Periodically determine serum urea nitrogen (BUN), serum creatinine, or creatinine clearance (CrCl). When feasible, it is recommended that serial audiograms be obtained in patients old enough to be tested, particularly high-risk patients. Evidence of ototoxicity (dizziness, vertigo, tinnitus, roaring in the ears, or hearing loss) or nephrotoxicity requires dosage adjustment or discontinuance of the drug. As with the other aminoglycosides, on

rare occasions, changes in renal and eighth cranial nerve function may not manifest until soon after completion of therapy.

Monitor serum concentrations of aminoglycosides when feasible to ensure adequate levels and to avoid potentially toxic levels. When monitoring gentamicin peak concentrations, adjust dosage so that prolonged levels above 12 mcg/mL are avoided. When monitoring gentamicin trough concentrations, adjust dosage so that levels above 2 mcg/mL are avoided. Excessive peak or trough serum concentrations of aminoglycosides may increase the risk of renal and eighth cranial nerve toxicity. In the event of overdose or toxic reactions, hemodialysis may aid in the removal of gentamicin from the blood, especially if renal function is, or becomes, compromised. The rate of removal of gentamicin is considerably less by peritoneal dialysis than by hemodialysis.

Avoid concurrent and/or sequential systemic or topical use of other potentially neurotoxic and/or nephrotoxic drugs, such as cisplatin, cephaloridine, kanamycin, amikacin, neomycin, polymyxin B, colistin, paromomycin, streptomycin, tobramycin, vancomycin, and viomycin. Other factors that may increase patient risk of toxicity are advanced age and dehydration.

Avoid the concurrent use of gentamicin with potent diuretics, such as ethacrynic acid or furosemide, since certain diuretics by themselves may cause ototoxicity. In addition, when administered intravenously (IV), diuretics may enhance aminoglycoside toxicity by altering the antibiotic concentration in serum and tissue.

Pregnancy:

Aminoglycosides can cause fetal harm when administered to a pregnant woman.

Pharmacologic Category

Antibiotic, Aminoglycoside

Dosing: Adult

Dosage guidance:

Dosing: For patients with total body weight (TBW) < ideal body weight (IBW), calculate the dose based on TBW. For patients with TBW 1 to 1.2 times IBW, calculate the dose based on TBW or IBW. TBW may be preferred in patients who may have increased V_d (eg, critically ill). For patients with TBW >1.2 times IBW, use adjusted body weight

$([0.4 \times \{TBW - IBW\}] + IBW)$ for initial weight-based dosing and for estimating kidney function with Cockcroft-Gault (CrCl) ([Ref](#)).

Clinical considerations: Gentamicin is not recommended for the treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)). Therapeutic drug monitoring is recommended to ensure efficacy and avoid toxicity, particularly in patients who are critically ill with serious infections or in disease states known to significantly alter aminoglycoside pharmacokinetics (eg, cystic fibrosis, burns, major surgery). Timing and frequency of concentration monitoring is individualized based on dosing and monitoring strategy ([Ref](#)).

Usual dosage range:

Gram-negative infections:

Conventional/traditional dosing: **IV, IM:** 3 to 5 mg/kg/day in divided doses every 8 hours ([Ref](#)). Some experts favor an initial loading dose of 2.5 to 3 mg/kg ([Ref](#)). Target peak concentration depends on indication and site of infection; in general, adjust dose to achieve peak of 7 to 10 mcg/mL for serious infections (including life-threatening infections). Target trough concentrations should be <2 mcg/mL; ideal target <1 mcg/mL ([Ref](#)).

High-dose extended-interval dosing (once-daily dosing): **IV:** 5 to 7 mg/kg once daily ([Ref](#)). Adjust gentamicin dose and interval to achieve an extrapolated peak concentration of ~15 to 20 mcg/mL and trough concentration ≤ 0.5 mcg/mL ([Ref](#)). **Note:** Published nomograms for dosage adjustment may not apply to patients with altered pharmacokinetics (eg, patients with ascites, burns covering >20% total BSA, end-stage renal disease requiring hemodialysis, pregnancy) ([Ref](#)); some experts prefer traditional intermittent dosing in patients with impaired (ie, CrCl <30 mL/minute) or unstable kidney function ([Ref](#)).

Synergy dosing for non-CNS gram-positive infections:

IV, IM: 3 mg/kg/day in 1 to 3 divided doses in combination with a gram-positive active agent ([Ref](#)). When divided doses are used, adjust gentamicin dose to achieve peak concentration of 3 to 4 mcg/mL and trough concentration <1 mcg/mL ([Ref](#)).

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***Bartonella* spp. infection**



***Bartonella* spp. infection (off-label use):**

Bacteremia with or without endocarditis: IV: 3 mg/kg/day in 1 or 3 divided doses in combination with doxycycline for 14 days, followed by doxycycline monotherapy ([Ref](#)). For patients with HIV and bacteremia without endocarditis, some experts do not recommend gentamicin as part of the regimen ([Ref](#)).

Cat scratch disease, disseminated (eg, hepatosplenic, prolonged systemic febrile illness) (alternative agent): IV: 3 mg/kg once daily in combination with rifampin for 10 to 14 days ([Ref](#)).

Note: In contrast to other gram-negative infections, target a gentamicin peak concentration of 3 to 4 mcg/mL and trough concentration of <1 mcg/mL when divided doses are used ([Ref](#)).

Bloodstream infection

Bloodstream infection:

Note: For adjunctive empiric therapy for patients at high risk for resistant gram-negative pathogens ([Ref](#)); however, some experts recommend other aminoglycosides in this setting ([Ref](#)); gentamicin is not recommended for treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)).

IV: 5 to 7 mg/kg once daily as part of an appropriate combination regimen; once culture and susceptibility results are available, can generally discontinue and use a single agent with documented activity. Gentamicin is not recommended as monotherapy ([Ref](#)).

Antibiotic lock technique (catheter-salvage strategy) (off-label use): **Note:** For infections caused by susceptible organisms when the catheter cannot be removed; use in addition to systemic antibiotics ([Ref](#)).

Intracatheter: Prepare lock solution to final concentration of gentamicin 1 to 5 mg/mL (may be combined with heparin) ([Ref](#)). The gentamicin concentration may vary by institution, catheter type, and whether heparin is utilized; solutions with heparin are preferred. Prepare solution immediately before instillation with a sufficient volume to fill the catheter (2 to 5 mL). Instill into each lumen of the catheter access port with a dwell time of up to 72 hours, depending on frequency of catheter use. Withdraw lock solution prior to catheter use; replace with fresh gentamicin lock solution after catheter use. Antibiotic lock therapy is given for the same duration as systemic antibiotics ([Ref](#)).

Brucellosis



Brucellosis (off-label use):

Note: Additional agents or other regimens are preferred for neurobrucellosis and infection in patients who are pregnant ([Ref](#)).

Endocarditis: IV, IM: 5 mg/kg/day in 1 to 3 divided doses for 4 weeks as part of an appropriate combination regimen ([Ref](#)).

Spondylitis: IV, IM: 5 mg/kg once daily for 7 to 14 days as part of an appropriate combination regimen ([Ref](#)).

Uncomplicated (nonfocal): IV, IM: 5 mg/kg once daily for 7 to 10 days as part of an appropriate combination regimen ([Ref](#)).

CNS infection



CNS infection

Cerebrospinal fluid shunt infection, health care-associated:

Note: Intraventricular/intrathecal administration is generally reserved for infections caused by extensively drug-resistant organisms, infections refractory to appropriate parenteral therapy, or when infected shunts cannot be removed ([Ref](#)). Gentamicin is not recommended for treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)).

Intraventricular (adjunct to systemic therapy; use a preservative-free preparation):
Initial: 4 to 8 mg/day; some experts recommend adjusting dosage and administration interval based on cerebrospinal fluid gentamicin concentrations (goal: 10 to 20 times MIC of causative organism), ventricle size, and daily output from ventricular drain ([Ref](#)). When intraventricular gentamicin is administered via a ventricular drain, clamp drain for 15 to 60 minutes after administration (allows solution to equilibrate in cerebrospinal fluid). Duration is individualized according to clinical and microbiologic response ([Ref](#)).

Meningitis, bacterial:

Enterococcus spp.: IV: 5 mg/kg/day in 3 divided doses ([Ref](#)); give as part of an appropriate combination regimen and individualize duration based on clinical response ([Ref](#)).

Listeria monocytogenes: **IV**: 5 mg/kg/day in 3 divided doses in combination with ampicillin or penicillin. Gentamicin is given until clinical improvement (typically ≥ 7 days) or in poor responders for up to 21 days, as long as there are no signs of nephrotoxicity or ototoxicity; total duration of antibiotic therapy is ≥ 21 days ([Ref](#)).

Endocarditis, treatment

Endocarditis, treatment:

***Enterococcus* spp. (native or prosthetic valve, without high-level gentamicin resistance):** **IV, IM**: 1 mg/kg every 8 hours **or** 3 mg/kg once daily as part of an appropriate combination regimen. Duration recommendations vary; the American Heart Association (AHA) recommends 4 to 6 weeks depending on duration of symptoms prior to presentation, source of infection, and the specific combination regimen, while the European Society of Cardiology (ESC) recommends gentamicin for the first 2 weeks of a 6-week regimen ([Ref](#)). **Note:** For native-valve endocarditis due to ampicillin-susceptible *Enterococcus faecalis*, some experts prefer a combination regimen that does not contain gentamicin ([Ref](#)).

***Staphylococcus* spp. (prosthetic valve) (off-label use):** **IV, IM**: 3 mg/kg/day in 1 to 3 divided doses for the first 2 weeks of an appropriate combination regimen; continue the other components for a total of ≥ 6 weeks ([Ref](#)).

***Viridans* group streptococci and *Streptococcus bovis* (off-label use):**

Native valve: Highly penicillin-susceptible (MIC ≤ 0.12 mcg/mL): **IV, IM**: 3 mg/kg/day once daily (preferred) or in 3 divided doses (alternative) in combination with a beta-lactam for 2 weeks. **Note:** This regimen is reserved for patients with uncomplicated infection, prompt response to therapy, and no preexisting renal failure or cranial nerve VIII disease ([Ref](#)).

Native valve: Relatively penicillin-resistant (MIC > 0.12 and < 0.5 mcg/mL): **IV, IM**: 3 mg/kg/day once daily (preferred) or in 3 divided doses (alternative) in combination with a beta-lactam. Gentamicin duration is for the first 2 weeks of the total 4-week regimen ([Ref](#)).

Native valve: Penicillin-resistant (MIC ≥ 0.5 mcg/mL): **IV, IM**: 3 mg/kg/day once daily (preferred) or in 3 divided doses (alternative) in combination with a beta-lactam ([Ref](#)). Gentamicin duration is for the first 2 weeks of the total 4-week regimen ([Ref](#)).

Prosthetic valve: Highly penicillin-susceptible ($\text{MIC} \leq 0.12$ mcg/mL): **IV, IM:** 3 mg/kg/day once daily (preferred) or in 3 divided doses (alternative) in combination with a beta-lactam. Gentamicin duration is for the first 2 weeks of the total 6-week regimen ([Ref](#)).

Prosthetic valve: Relatively penicillin-resistant ($\text{MIC} > 0.12$ and < 0.5 mcg/mL) or fully penicillin-resistant ($\text{MIC} \geq 0.5$ mcg/mL): **IV, IM:** 3 mg/kg once daily in combination with a beta-lactam for 6 weeks ([Ref](#)); some experts prefer shorter courses of the gentamicin component (≥ 2 weeks) ([Ref](#)).

Osteomyelitis, prevention, following open fractures ^

Osteomyelitis, prevention, following open fractures (type III [severe contamination or comminution]) (off-label use): **IV:** 5 to 6 mg/kg every 24 hours as part of an appropriate combination regimen. Duration is 72 hours after injury or up to 24 hours after wound closure ([Ref](#)).

Pelvic infections ^

Pelvic infections (off-label use):

Note: Dose is based on actual body weight. For patients > 1.2 times IBW, use adjusted body weight ($\text{IBW} + [0.4 \times (\text{TBW} - \text{IBW})]$) ([Ref](#)).

Intra-amniotic infection (chorioamnionitis) : **IV:** 5 mg/kg once every 24 hours (preferred) **or** 2 mg/kg loading dose followed by 1.5 mg/kg every 8 hours (alternative); give in combination with ampicillin ([Ref](#)). In patients undergoing cesarean delivery, add anaerobic coverage (eg, clindamycin or metronidazole) ([Ref](#)). Some experts add anaerobic coverage regardless of delivery route and add a macrolide for genital mycoplasmas ([Ref](#)).

Duration: Continue regimen until vaginal delivery or for 1 dose after cesarean delivery ([Ref](#)). Some experts recommend 1 additional dose after vaginal delivery and extension of antibiotics after cesarean delivery until patient is afebrile and asymptomatic ≥ 48 hours ([Ref](#)).

Postpartum endometritis: **IV:** 5 mg/kg once daily (preferred) **or** 1.5 mg/kg every 8 hours (alternative) in combination with clindamycin. Treat until the patient is clinically improved (no fundal tenderness) and afebrile for 24 to 48 hours ([Ref](#)).

Note: For patients with group B *Streptococcus* colonization or sepsis, additional agents or other regimens are preferred ([Ref](#)).

Peritonitis, treatment, peritoneal dialysis

Peritonitis, treatment, peritoneal dialysis (off-label route):

Note: Intraperitoneal administration is preferred to IV administration unless the patient has sepsis. For empiric therapy, use as part of an appropriate combination regimen. Continuous intraperitoneal administration of aminoglycosides is not recommended ([Ref](#)). Gentamicin is not recommended for the treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)).

Intermittent: Intraperitoneal: 0.6 mg/kg added to the dialysate solution once daily; allow to dwell ≥ 6 hours. If cultures are negative and patient is improving, can generally discontinue gentamicin after 3 days ([Ref](#)). If cultures are positive, if possible, switch to another active antibiotic class to decrease the risk of toxicity; otherwise, for patients with adequate clinical response, duration of therapy is ≥ 3 weeks. Prolonged courses (> 7 days) of gentamicin are generally not recommended. For patients with no improvement after 5 days, remove catheter and treat with appropriate systemic antibiotics for 14 days after catheter removal ([Ref](#)).

Plague

Plague (*Yersinia pestis*), treatment (off-label use):

Note: Consult public health officials for event-specific recommendations.

IV, IM: 5 mg/kg once daily for 7 to 14 days and for at least a few days after clinical resolution ([Ref](#)).

Sepsis or septic shock

Sepsis or septic shock:

Note: For adjunctive empiric therapy for patients at high risk for multidrug resistant gram-negative pathogens ([Ref](#)); however, some experts recommend other aminoglycosides in this setting ([Ref](#)); gentamicin is not recommended for treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)).

IV: 5 to 7 mg/kg once daily in combination with a second gram-negative agent ([Ref](#)); once culture and susceptibility tests are available, can generally

discontinue and use a single agent with documented activity. Gentamicin is not recommended as monotherapy ([Ref](#)).

Sexually transmitted infections

Sexually transmitted infections:

Gonococcal infection, uncomplicated (infection of the cervix, rectum, or urethra) (alternative agent) (off-label use):

Note: Reserve for patients who cannot use a cephalosporin ([Ref](#)).

>45 kg: **IM:** 240 mg (administer as two 3 mL injections) once ([Ref](#)).

≤45 kg: **IM:** 5 mg/kg once ([Ref](#)).

Administer in combination with oral azithromycin 2 g, each as a single dose ([Ref](#)). When treatment failure is suspected (eg, detection of *Neisseria gonorrhoeae* after treatment without additional sexual exposure), consult an infectious diseases specialist. Report failures to the CDC through state and local health departments ([Ref](#)).

Pelvic inflammatory disease, severe (including tubo-ovarian abscess) (alternative agent) (off-label use): **IV, IM:** 3 to 5 mg/kg once daily **or** 2 mg/kg as a loading dose followed by 1.5 mg/kg every 8 hours; give as part of an appropriate combination regimen; after 24 to 48 hours of sustained clinical improvement, transition to oral therapy to complete 14 days of treatment ([Ref](#)). **Note:** Some experts reserve this regimen for patients who cannot use preferred agents due to greater associated adverse effects ([Ref](#)).

Surgical prophylaxis

Surgical prophylaxis (alternative agent for select GI tract, GU tract, or gynecologic/obstetric procedures) (off-label use):

IV: 5 mg/kg as a single dose within 60 minutes prior to surgical incision; give in combination with other antibiotics (procedure dependent) ([Ref](#)). **Note:** In cases where extension of prophylaxis is warranted postoperatively, total duration should be ≤24 hours ([Ref](#)). Postoperative prophylaxis is not recommended in clean and clean-contaminated surgeries ([Ref](#)).

Tularemia

Tularemia (*Francisella tularensis*) (off-label use):

Note: For severe infection, may consider use as part of an appropriate combination regimen. For neuroinvasive tularemia, combination therapy is recommended ([Ref](#)).

IV, IM: 6 mg/kg once daily; duration is generally 10 days ([Ref](#)).

Urinary tract infection **Urinary tract infection (alternative agent):**

Note: Reserve for patients with or at risk for resistant gram-negative infection ([Ref](#)). Gentamicin is not recommended for the treatment of *P. aeruginosa* due to insufficient target attainment ([Ref](#)).

Cystitis, acute uncomplicated or acute simple cystitis (infection limited to bladder without signs/symptoms of upper tract, prostate, or systemic infection): **IV:** 5 mg/kg once ([Ref](#)).

Urinary tract infection, complicated (including pyelonephritis or urinary tract infection with systemic signs/symptoms):

IV, IM: 5 to 7 mg/kg once; subsequent doses/dosing interval based on pharmacokinetic evaluation ([Ref](#)). For patients with symptomatic improvement within the first ~48 hours of therapy, total duration (including oral step-down therapy if appropriate) is 5 to 7 days (7 days if therapy is completed with gentamicin) ([Ref](#)).

Pregnant patients: **IV:** 1.5 mg/kg every 8 hours **or** 5 mg/kg every 24 hours. Administer in combination with IV ampicillin. Switch to an appropriate oral regimen once patient has improvement in symptoms. Total duration of therapy is 14 days ([Ref](#)); for patients with symptomatic improvement within the first 48 hours of therapy, some experts recommend shorter courses of 7 to 10 days ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function:

High-dose, extended-interval dosing:

Note: Use with caution in patients with CrCl <40 mL/minute ([Ref](#)); high-dose, extended-interval dosing may still be considered, especially in patients with severe sepsis/shock or those infected with multidrug-resistant gram-negative organisms ([Ref](#)).

IV: Initial dose: 5 to 7 mg/kg. Subsequent doses and frequency of administration should be determined based on therapeutic drug monitoring; regimens may vary; nomograms exist to guide dose adjustments, although individualized calculations may be necessary in patients with highly variable or altered aminoglycoside pharmacokinetics (eg, critical illness, pregnancy) ([Ref](#)). Also refer to institutional-specific policies. The following recommendations may serve as a general guideline after the initial dose:

CrCl $\geq$ 60 mL/minute: **IV:** Administer every 24 hours; adjust dose and/or interval based on gentamicin serum concentrations.

CrCl 40 to <60 mL/minute: **IV:** Administer every 36 hours; adjust dose and/or interval based on gentamicin serum concentrations.

CrCl 20 to <40 mL/minute: **IV:** Administer every 48 hours; adjust dose and/or interval based on gentamicin serum concentrations.

CrCl <20 mL/minute: **IV:** Administer usual dose once, then determine subsequent dose and interval based on serum concentration monitoring. Some published protocols would recommend conventional/traditional dosing in these patients ([Ref](#)).

Conventional/traditional dosing:

Note: High-dose, extended-interval dosing is generally preferred for treatment of gram-negative infections. Conventional/traditional dosing is typically used for synergy dosing or non-CNS, gram-positive infections.

Regimens may vary based on individualized pharmacokinetic calculations and pharmacodynamic targets; also refer to institutional-specific policies. **Note:** The following recommendations are expert opinion derived from Leroy 1978 and based on a usual dosage range of 3 to 5 mg/kg/day:

CrCl ≥ 60 mL/minute: **IM, IV:** No dosage adjustment necessary.

CrCl ≥ 40 to < 60 mL/minute: **IM, IV:** Administer usual dose every 12 hours; adjust dose and/or interval based on gentamicin serum concentrations.

CrCl 20 to < 40 mL/minute: **IM, IV:** Administer usual dose every 24 hours; adjust dose and/or interval based on gentamicin serum concentrations.

CrCl < 20 mL/minute: **IM, IV:** Administer usual dose every 36 to 48 hours; adjust dose and/or interval based on gentamicin serum concentrations.

Augmented renal clearance (measured urinary CrCl ≥ 130 mL/minute/1.73 m²):

Note: Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Young patients (< 55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients ([Ref](#)).

High-dose, extended-interval dosing: **IV:** Initial: 7 to 8 mg/kg once daily for known/suspected sepsis; adjust dose and/or interval based on close monitoring of gentamicin serum concentrations and individualized pharmacokinetic calculations and pharmacodynamic targets; use of nomograms for dosage adjustment are not recommended in this population ([Ref](#)).

Hemodialysis, intermittent (thrice weekly): Dialyzable ($\sim 50\%$; dependent on filter and duration): **Note:** Postdialysis concentrations should be drawn ≥ 2 and up to 4 hours after hemodialysis (HD) to allow for redistribution ([Ref](#)).

Loading dose of 2 to 3 mg/kg followed by:

Gram-positive synergy dosing: **IM, IV:** 1 mg/kg/dose 3 times weekly after dialysis on dialysis days ([Ref](#)); consider redosing for pre-HD or post-HD concentrations < 1 mg/L.

Urinary tract infection: **IM, IV:** 1 to 1.5 mg/kg/dose 3 times weekly after dialysis on dialysis days ([Ref](#)); consider redosing for pre-HD concentrations < 1.5 to 2 mg/L or post-HD concentrations < 1 mg/L.

Systemic gram-negative infection: **IM, IV:** 2 to 3 mg/kg/dose 3 times weekly after dialysis on dialysis days ([Ref](#)); consider redosing for pre-HD concentrations < 3 to 5 mg/L or post-HD concentrations < 2 mg/L.

Peritoneal dialysis: **IM, IV:** Initial: 1 to 3 mg/kg/dose (depending on infection site, severity, and susceptibility of infecting organisms) every 48 to 72 hours based on gentamicin

serum concentrations ([Ref](#)).

CRRT:

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection) and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, nephrotoxicity) due to drug accumulation is important.

CVVH/CVVHD/CVVHDF:

Gram-positive synergy dosing: **IV:** Loading dose of 2 to 3 mg/kg/dose followed by 1 to 1.5 mg/kg every 24 to 36 hours (redose when concentration <1 mg/L ([Ref](#))).

Urinary tract infection: **IV:** Loading dose of 2 to 3 mg/kg/dose followed by 1 to 1.5 mg/kg every 24 to 36 hours (redose when concentration <1.5 to 2 mg/L ([Ref](#))).

Systemic gram-negative infection: **IV:** Initial: 2.5 to 3 mg/kg/dose; up to 5 mg/kg/dose may be considered (depending on infection site, severity, and susceptibility of infecting organisms) every 24 to 48 hours based on gentamicin serum concentrations (expert opinion derived from Petejova 2012).

PIRRT (eg, sustained, low-efficiency diafiltration):

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection) and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, nephrotoxicity) due to drug accumulation is important. Dosing below assumes daily use of PIRRT.

Gram-positive synergy dosing: **IV:** Loading dose of 2 to 3 mg/kg/dose followed by 1 to 1.5 mg/kg/dose every 24 to 36 hours; adjust dose and/or frequency based on gentamicin serum concentrations. When scheduled dose falls on a PIRRT day, administer after PIRRT ([Ref](#)).

Urinary tract infection: **IV:** Loading dose of 2 to 3 mg/kg/dose followed by 1 to 1.5 mg/kg/dose every 24 to 36 hours; adjust dose based on serum concentrations. When scheduled dose falls on a PIRRT day, administer after PIRRT ([Ref](#)).

Systemic gram-negative infection: **IV:** Initial: 6 mg/kg/dose every 48 hours; adjust interval if needed based on gentamicin levels (usual frequency is every 48 to 72 hours). **Note:** Administer each dose 30 to 60 minutes prior to PIRRT session on PIRRT days ([Ref](#)).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; however dosage adjustment is not likely to be necessary (does not undergo hepatic metabolism).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

IV: Use **adjusted body weight** for initial weight-based dosing when targeting $C_{\max}$ /minimum inhibitory concentration (MIC) goals with either loading dose equation (based on target aminoglycoside concentration and estimated V_d) or mg/kg approach and when estimating kidney function with Cockcroft-Gault (CrCl) ([Ref](#)). Alternatively, use **adjusted body weight** to estimate CrCl and aminoglycoside dosing with the Bayesian approach when targeting AUC/MIC goals ([Ref](#)). **Note:** If aminoglycoside therapy is continued, use $C_{\max}$ /MIC or AUC/MIC goals to optimize therapy, especially in the critically ill where weight and kidney function may be poor surrogates of V_d and clearance ([Ref](#)).

Rationale for recommendations: Aminoglycosides are hydrophilic medications with a low V_d and clearance that is proportional to GFR. Pharmacokinetic studies have observed adjusted body weight, using a correction factor of 0.4, is the most appropriate weight metric to correct V_d in the setting of obesity. However, there is wide variation in the correction factors reported that could lead to under- or overdosing in clinical practice ([Ref](#)). Early use of therapeutic drug monitoring is recommended ([Ref](#)).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "[Gentamicin \(systemic\): Pediatric drug information](#)")

Dosage guidance:

Dosing: Initial dosing recommendations presented. Monitoring of serum concentrations is recommended to ensure efficacy and avoid toxicity. Timing and frequency of concentration monitoring is individualized based on dosing and monitoring strategy ([Ref](#)). Routes of administration may vary (including IM, IV, intraperitoneal, intrathecal, and intraventricular); use caution. Some dosing is based on tobramycin studies.

Dosing consideration for obesity: In pediatric patients who are obese, use adjusted body weight ($IBW + 0.4 [TBW - IBW]$) to calculate initial dosage ([Ref](#)). Alternatively, adjusted body weight for pediatric patients who are obese may be calculated using the equation $0.7 \times TBW$ ([Ref](#)), or fat-free mass can be used to calculate the initial dose in pediatric patients ≥ 2 years regardless of body habitus ([Ref](#)). Dosage should then be individualized based upon serum concentration monitoring.

General dosing, susceptible infection: Note: Optimal dose and frequency not established in patients receiving extracorporeal membrane oxygenation (ECMO); patient-specific considerations (eg, reason for ECMO) and variability with ECMO procedure itself make extrapolation of pharmacokinetic data and dosing to all patients receiving ECMO difficult; closely monitor serum concentrations and determine individual dosing needs in these patients.

Conventional dosing: Infants, Children, and Adolescents: IM, IV: 2 to 2.5 mg/kg/dose every 8 hours ([Ref](#)).

Extended-interval dosing: Limited data available:

Weight-directed: Infants, Children, and Adolescents: IV: 5 to 7.5 mg/kg/dose every 24 hours ([Ref](#)).

Age-directed: Based on data from 114 patients, the following has been suggested ([Ref](#)):

Infants and Children ≥ 3 months to < 2 years: IV: 9.5 mg/kg/dose every 24 hours.

Children 2 to < 8 years: IV: 8.5 mg/kg/dose every 24 hours.

Children ≥ 8 years and Adolescents: IV: 7 mg/kg/dose every 24 hours.

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Brucellosis

Brucellosis: Limited data available: Infants, Children, and Adolescents: IV, IM: 5 mg/kg/dose once daily as part of an appropriate combination regimen ([Ref](#)). **Note:** Duration depends on extent of disease; when gentamicin is included in regimen, it is typically given for the first 7 to 10 days of therapy ([Ref](#)).

CNS infection

CNS infection:

Meningitis, including health care-associated meningitis: Infants, Children, and Adolescents: IV: 7.5 mg/kg/**day** divided every 8 hours in combination with

other antibiotics; duration should be individualized based on patient characteristics, infecting organism, and response ([Ref](#)).

Ventriculitis, including health care-associated ventriculitis and cerebrospinal fluid (CSF) shunt infections: Limited data available; optimal dose not established ([Ref](#)).

Dosage and administration interval can be adjusted based on CSF gentamicin concentrations, ventricle size, and daily output from ventricular drain.

Duration is individualized according to clinical and microbiological response ([Ref](#)).

Intraventricular/intrathecal (use a preservative-free preparation):

Infants and Children: 1 to 2 mg/**day**.

Adolescents: Dosing recommendations not reported; however, in adults, 4 to 8 mg/**day** has been suggested.

Cystic fibrosis, pulmonary infection

Cystic fibrosis, pulmonary infection: Infants, Children, and Adolescents:

Conventional dosing: IM, IV: 3.3 mg/kg/dose every 8 hours ([Ref](#)).

Extended-interval dosing: IV: 10 to 12 mg/kg/dose every 24 hours ([Ref](#)); maximum reported dose from a survey of 28 Cystic Fibrosis (CF) Foundation-accredited centers ranged from 12 to 20 mg/kg/dose ([Ref](#)). **Note:** The CF Foundation recommends extended-interval dosing as preferred over conventional dosing.

Duration of therapy: Treatment duration varies and is dependent on patient-specific factors including response to therapy. Typical duration is 10 to 14 days; some patients may require extended therapy (eg, 21 days) ([Ref](#)).

Endocarditis, treatment

Endocarditis, treatment:

Synergy dosing (eg, gram-positive bacteria): Children and Adolescents: IV: 3 to 6 mg/kg/**day** divided every 8 hours ([Ref](#)); 3 mg/kg/**day** may also be administered every 24 hours for some pathogens; see guidelines for details ([Ref](#)). Use in combination with other antibiotics dependent on pathogen and source of infection (ie, valve type) ([Ref](#)).

Treatment dosing (eg, gram-negative bacteria): Children and Adolescents: IV: 7.5 mg/kg/**day** divided every 8 hours; use in combination with other antibiotics

(Ref).

Gonococcal infection, uncomplicated infection of the cervix, urethra, or rectum

Gonococcal infection, uncomplicated infection of the cervix, urethra, or rectum (alternative agent): Limited data available:

Note: Reserve for patients who cannot receive first-line therapies (eg, ceftriaxone); report treatment failures to the CDC through state and local health departments (Ref).

Adolescents ≥ 15 years:

≤ 45 kg: IM: 5 mg/kg once in combination with a single dose of oral azithromycin (Ref).

> 45 kg: IM: 240 mg once (administer as two 3 mL injections) in combination with a single dose of oral azithromycin (Ref).

Intra-abdominal infection, complicated

Intra-abdominal infection, complicated: Infants, Children, and Adolescents: IV: 3 to 7.5 mg/kg/day divided every 8 to 24 hours; use in combination with other antibiotics (Ref).

Peritonitis

Peritonitis (peritoneal dialysis): Limited data available: Infants, Children, and Adolescents: Intraperitoneal: Intermittent: 0.6 mg/kg/dose every 24 hours in the long dwell (Ref).

Surgical prophylaxis

Surgical prophylaxis: Infants, Children, and Adolescents: IV: 2.5 mg/kg as a single dose; administer within 60 minutes prior to surgical incision with or without other antibiotics (procedure dependent) (Ref).

Tularemia

Tularemia:

Infants, Children, and Adolescents: IV, IM: 5 to 7.5 mg/kg/dose every 24 hours; may increase to 9.5 mg/kg/dose based on age and AUC monitoring (specific target not provided in guidelines) ([Ref](#)). Consider dosing based on age as follows:

≤10 years: IV, IM: 7.5 mg/kg/dose every 24 hours ([Ref](#)).

>10 years: IV, IM: 6 mg/kg/dose every 24 hours ([Ref](#)).

Duration of therapy: Typical treatment duration is 10 days total; may transition to an oral regimen following defervescence and clinical improvement ([Ref](#)); some suggest 7 days of therapy may be adequate for mild disease ([Ref](#)). Longer duration is needed for more severe illness including meningitis ([Ref](#)); for neuroinvasive disease, while gentamicin is typically discontinued after 10 days, total duration of therapy should be 21 days ([Ref](#)).

Urinary tract infection



Urinary tract infection (UTI):

Note: Duration of therapy for multiple-dose regimens: For uncomplicated cystitis in patients ≥3 months of age, treat for 3 to 5 days; patients <2 years of age may require a longer course (eg, 7 days). For complicated UTI, including pyelonephritis, treat for a total of 6 to 10 days; while 7 to 14 days has been recommended for complicated UTI, this was not shown to improve outcomes compared to a shorter duration of 6 to 10 days ([Ref](#)).

Conventional dosing: Infants, Children, and Adolescents: IV: 7.5 mg/kg/**day** divided every 8 hours until clinical improvement and able to tolerate oral intake; complete course with oral antibiotics. Duration should be individualized based upon age, severity, and degree of urinary tract involvement.

Extended-interval dosing: Limited data available: Based on data from 90 patients (ages: 1 month to 12 years), the following age-directed dosing has been suggested ([Ref](#)): **Note:** Patients were transitioned to oral therapy once afebrile for 24 hours.

Infants and Children <5 years: IV: 7.5 mg/kg/dose every 24 hours.

Children 5 to 10 years: IV: 6 mg/kg/dose every 24 hours.

Children 11 to 12 years: IV: 4.5 mg/kg/dose every 24 hours.

Single-dose regimen: Limited data available: **Note:** Recommended for treatment of uncomplicated cystitis caused by antimicrobial resistant gram-negative

pathogens; guidelines do not address pediatric dosing but an IV single dose is an option for adults ([Ref](#)):

Infants, Children, and Adolescents: IM: 5 mg/kg as a single dose; dosing based on 2 prospective studies evaluating single-dose IM gentamicin in patients 1 month to 15 years and a systematic review evaluating studies of various single-dose aminoglycosides in pediatric and adult patients ([Ref](#)); one study limited doses to 300 mg ([Ref](#)). An overall pooled cure rate for single dose IM aminoglycoside for treatment of children and adults with mainly uncomplicated cystitis was reported as $94.5\% \pm 4.3\%$ ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Altered kidney function:

Note: Gentamicin serum concentrations should be monitored in patients with kidney impairment; following the initial dose, subsequent doses may be determined based on therapeutic monitoring.

Infants, Children, and Adolescents: IM, IV:

The following adjustments have been recommended ([Ref](#)): **Note:** Renally adjusted dose recommendations are based on doses of 2.5 mg/kg/dose every 8 hours:

GFR >50 mL/minute/1.73 m²: No dosage adjustment necessary.

GFR 30 to 50 mL/minute/1.73 m²: Administer every 12 to 18 hours.

GFR 10 to 29 mL/minute/1.73 m²: Administer every 18 to 24 hours.

GFR <10 mL/minute/1.73 m²: Administer every 48 to 72 hours.

Intermittent hemodialysis: 2 mg/kg/dose; redose as indicated by serum concentration.

Peritoneal dialysis (PD): 2 mg/kg/dose; redose as indicated by serum concentration.

Continuous renal replacement therapy (CRRT): 2 to 2.5 mg/kg/dose every 12 to 24 hours, monitor serum concentrations.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; however, dosage adjustment not likely to be necessary (does not undergo hepatic metabolism).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Frequency not defined.

Cardiovascular: Edema, hypertension, hypotension, phlebitis, thrombophlebitis

Central nervous system: Abnormal gait, ataxia, brain disease, confusion, depression, dizziness, drowsiness, headache, lethargy, myasthenia, numbness, paresthesia, peripheral neuropathy, pseudomotor cerebri, seizure, vertigo

Dermatologic: Alopecia, erythema, pruritus, skin rash, urticaria

Endocrine & metabolic: Hypocalcemia, hypokalemia, hypomagnesemia, hyponatremia, weight loss

Gastrointestinal: Anorexia, *Clostridioides difficile*-associated diarrhea, decreased appetite, enterocolitis, nausea, sialorrhea, stomatitis, vomiting

Genitourinary: Casts in urine (hyaline, granular), Fanconi-like syndrome (infants and adults; high dose, prolonged course), oliguria, proteinuria

Hematologic & oncologic: Agranulocytosis, anemia, eosinophilia, granulocytopenia, leukopenia, purpura, reticulocytopenia, reticulocytosis, splenomegaly, thrombocytopenia

Hepatic: Hepatomegaly, increased liver enzymes

Hypersensitivity: Anaphylaxis, anaphylactoid reaction, hypersensitivity reaction

Local: Injection site reaction, pain at injection site

Neuromuscular & skeletal: Arthralgia, muscle cramps, muscle fatigue (myasthenia gravis-like syndrome), muscle twitching, tremor, weakness

Ophthalmic: Visual disturbance

Otic: Auditory impairment, hearing loss (associated with persistently increased serum concentrations; early toxicity usually affects high-pitched sound), tinnitus

Renal: Decreased creatinine clearance, decreased urine specific gravity, increased blood urea nitrogen, increased serum creatinine, polyuria, renal failure (high trough serum concentrations), renal tubular necrosis

Respiratory: Dyspnea, laryngeal edema, pulmonary fibrosis, respiratory depression

Miscellaneous: Fever

Contraindications

Hypersensitivity to gentamicin, other aminoglycosides, or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

- Hypersensitivity: Cross-sensitivity to other aminoglycosides may occur.
- Nephrotoxicity: **[US Boxed Warning]: May cause nephrotoxicity;** usual risk factors include preexisting renal impairment, concomitant nephrotoxic medications, advanced age and dehydration. Discontinue treatment if signs of nephrotoxicity occur; renal damage is usually reversible.
- Neuromuscular blockade and respiratory paralysis: May cause neuromuscular blockade and respiratory paralysis; especially when given soon after anesthesia or neuromuscular blockers.
- Neurotoxicity: **[US Boxed Warning]: May cause neurotoxicity;** usual risk factors include preexisting renal impairment, concomitant neuro-/nephrotoxic medications, advanced age and dehydration. Ototoxicity is proportional to the amount of drug given and the duration of treatment. Tinnitus or vertigo may be indications of vestibular injury and impending bilateral irreversible damage. Discontinue treatment if signs of ototoxicity occur.
- Superinfection: Prolonged use may result in fungal or bacterial superinfection, including *Clostridioides difficile*-associated diarrhea (CDAD) and pseudomembranous colitis; CDAD has been observed >2 months postantibiotic treatment.

Disease-related concerns:

- Electrolyte abnormalities: Use with caution in patients with hypocalcemia, hypokalemia, or hypomagnesemia.
- Hearing impairment: Use with caution in patients with preexisting vertigo, tinnitus, or hearing loss.
- Neuromuscular disorders: Use with caution in patients with neuromuscular disorders, including myasthenia gravis.
- Renal impairment: Use with caution in patients with preexisting renal insufficiency; dosage modification required.

Special populations:

- Patients with genomic variants in *MT-RNR1*: Carriers of certain variants in the *MT-RNR1* gene (eg, m.1555A>G) may be at increased risk for aminoglycoside-induced ototoxicity, including potentially significant hearing loss that may be irreversible, even when serum levels are within the normal range.
- Pregnancy: **[US Boxed Warning]: Aminoglycosides may cause fetal harm if administered to a pregnant woman.**

Concurrent drug therapy issues:

- Neurotoxic and/or nephrotoxic drugs: **[US Boxed Warning]: Avoid concomitant or sequential use of other neurotoxic and/or nephrotoxic drugs (eg, cisplatin, polymyxin B, colistin, vancomycin, other aminoglycosides).**
- Potent diuretics: **[US Boxed Warning]: Avoid concomitant use with potent diuretics (eg, ethacrynic acid, furosemide) since diuretics themselves may cause ototoxicity and may enhance aminoglycoside toxicity.**

Other warnings/precautions:

- Long-term use: Risk of toxicity is increased with extended duration of administration; additional monitoring may be required with long-term use.
- Surgical irrigation: May be almost completely systemically absorbed after local irrigation and/or topical application (except to the urinary bladder) during surgical procedures. Consider potential for nephrotoxicity, neuromuscular blockade, ototoxicity, and respiratory paralysis when administering aminoglycosides in this manner.

Warnings: Additional Pediatric Considerations

Use with caution in pediatric patients on extracorporeal membrane oxygenation (ECMO); pharmacokinetics of aminoglycosides may be altered; dosage adjustment and close monitoring necessary. Oral use for the prevention of NEC in premature neonates may potentially increase the risk for development of resistant bacteria; routine use for this is not recommended (Bury 2001; Reber 2004).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 40 mg/mL (2 mL, 20 mL)

Solution, Injection [preservative free]:

Generic: 10 mg/mL (2 mL); 40 mg/mL (2 mL, 20 mL)

Solution, Intravenous [preservative free]:

Generic: 60 mg (50 mL); 80 mg (50 mL, 100 mL); 100 mg (50 mL, 100 mL); 120 mg (100 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Gentamicin Sulfate Injection)

10 mg/mL (per mL): \$1.03 - \$4.22

40 mg/mL (per mL): \$0.73 - \$3.17

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 10 mg/mL (2 mL); 40 mg/mL (2 mL, 20 mL)

Solution, Intravenous:

Generic: 80 mg (50 mL); 100 mg (100 mL)

Administration: Adult

IM: Administer undiluted. Gentamicin in NS is not intended for IM administration.

IV: Infuse over 30 to 120 minutes.

Some penicillins (eg, carbenicillin, ticarcillin, piperacillin) have been shown to inactivate aminoglycosides in vitro. This has been observed to a greater extent with tobramycin and gentamicin, while amikacin has shown greater stability against inactivation. Concurrent use of these agents may pose a risk of reduced antibacterial efficacy in vivo, particularly in the setting of profound renal impairment. However, definitive clinical evidence is lacking. If combination penicillin/aminoglycoside therapy is desired in a patient with renal dysfunction, separation of doses (if feasible), and routine monitoring of aminoglycoside levels, CBC, and clinical response should be considered.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses. Initiate hyaluronidase antidote for refractory cases in addition to supportive management ([Ref](#)).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if infiltrated catheter remains in place, administer through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution ([Ref](#)).

Intraventricular (off-label route): Use preservative-free preparations only. When administered through a ventricular drain, clamp drain for 15 to 60 minutes before opening the drain to allow gentamicin solution to equilibrate in the cerebrospinal fluid ([Ref](#)).

Intraperitoneal (off-label route): May administer intermittently (1 exchange daily); allow to dwell ≥ 6 hours ([Ref](#)).

Preparation for Administration: Adult

IV: May dilute in NS or D5W. In adults, dilution in 50 to 200 mL is recommended; premix admixtures commercially available for some dosages.

Administration: Pediatric

Parenteral:

IM: May be administered undiluted. Appropriate site depends on age and development of muscle mass. In neonates and infants, IM injections are typically administered in the anterolateral thigh (vastus lateralis). For ages 1 to 2 years, the anterolateral thigh may be used or, if the muscle mass is adequately developed, the deltoid muscle. For ages ≥ 3 years, the deltoid muscle (for volumes typically less than ~1 mL) or anterolateral thigh may be used ([Ref](#)).

IV: Administer as diluted solution by slow intermittent infusion over 30 to 120 minutes; usual infusion time is 30 to 60 minutes; consider longer infusion time (60 to 120 minutes) with high doses. Shorter infusion times (≤ 5 minutes) have been reported in pediatric patients, including preterm and term neonates, receiving ≤ 4 mg/kg/dose ([Ref](#)). Avoid infusing concomitantly with penicillins or cephalosporins if feasible; consult drug interactions database for more information.

Gentamicin is a vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for antidote; elevate extremity; apply dry warm compresses. Hyaluronidase may be considered for refractory cases ([Ref](#)).

Intrathecal/intraventricular: Use preservative-free preparations only; must be diluted prior to administration. No specific administration information available; it has been suggested that instillation of small volumes (< 3 mL) over 1 to 2 minutes is safe ([Ref](#)). When administered through a ventricular drain, clamp drain for 15 to 60 minutes to allow gentamicin solution to equilibrate in the cerebrospinal fluid (CSF) ([Ref](#)).

Preparation for Administration: Pediatric

Parenteral:

IV: May dilute in NS or D5W. In adults, dilution in 50 to 200 mL is recommended; premix admixtures are commercially available for some dosages. In infants and children, the diluent volume should be less to allow for accurate measurement and administration. Maximum concentration is not addressed by the manufacturer; the concentration of the pediatric-specific product is 10 mg/mL; Vermont Oxford Network recommends a standard concentration of 2 mg/mL or 10 mg/mL for neonates ([Ref](#)); concentrations between 1 and 10 mg/mL have been reported by some institutions for use in neonatal and pediatric patients ([Ref](#)).

Intrathecal/intraventricular: Must use **preservative-free** gentamicin. Prior to administration dilute in **preservative-free** NS to a final concentration of 1 to 5 mg/mL ([Ref](#)).

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Store intact vials and premixed bags at 20°C to 25°C (68°F to 77°F). Protect from freezing. Stability information when diluted for infusion is not reported in the manufacturer's labeling;

refer to compatibility database for details.

Use: Labeled Indications

Serious infections: Treatment of serious infections (eg, sepsis, meningitis, urinary tract infections, respiratory tract infections, peritonitis, bone infections, skin and soft tissue infections) caused by susceptible strains of the following microorganisms: *Pseudomonas aeruginosa*, *Proteus* species (indole-positive and indole-negative), *Escherichia coli*, *Klebsiella* species, *Enterobacter* species, *Serratia* species, *Citrobacter* species, and *Staphylococcus* species (coagulase-positive and coagulase-negative); treatment of infective endocarditis caused by enterococci, in combination with other antibiotics.

Use: Off-Label: Adult

Bartonella spp. infections; Brucellosis; Chorioamnionitis; Endocarditis, treatment (viridans group streptococci and Streptococcus bovis [native or prosthetic valve]) (adults); Gonococcal infection, uncomplicated; Osteomyelitis, prevention, following open fractures (type III [severe contamination or comminution]); Pelvic inflammatory disease; Plague (Yersinia pestis), treatment; Postpartum endometritis; Surgical prophylaxis (alternative agent for select GI tract, GU tract, or gynecologic/obstetric procedures); Tularemia

Medication Safety Issues

[Collapse All](#)

Sound-alike/look-alike issues:

Gentamicin may be confused with gentian violet, kanamycin, vancomycin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a

complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Aescin: May increase nephrotoxic effects of Aminoglycosides. Management: Consider alternatives to this combination as aescin product labeling states coadministration with aminoglycosides is not recommended. *Risk D: Consider Therapy Modification*

Aminoglycosides: May increase nephrotoxic effects of Aminoglycosides. Aminoglycosides may increase neurotoxic effects of Aminoglycosides. *Risk X: Avoid*

Amphotericin B: May increase nephrotoxic effects of Aminoglycosides. Amphotericin B may increase neurotoxic effects of Aminoglycosides. *Risk C: Monitor*

Ataluren: May increase adverse/toxic effects of Aminoglycosides. Specifically, an increased risk of nephrotoxicity may occur with the concomitant use of ataluren and aminoglycosides. *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Intravesical): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Intravesical). *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Percutaneous): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Percutaneous). *Risk C: Monitor*

Bacillus clausii: Antibiotics may decrease therapeutic effects of Bacillus clausii. Management: Bacillus clausii should be taken in between antibiotic doses during concomitant therapy. *Risk D: Consider Therapy Modification*

Bacitracin (Systemic): May increase nephrotoxic effects of Aminoglycosides. Bacitracin (Systemic) may increase neurotoxic effects of Aminoglycosides. *Risk X: Avoid*

Bisphosphonate Derivatives: Aminoglycosides may increase hypocalcemic effects of Bisphosphonate Derivatives. Aminoglycosides may increase nephrotoxic effects of Bisphosphonate Derivatives. *Risk C: Monitor*

Botulinum Toxin-Containing Products: Aminoglycosides may increase neuromuscular-blocking effects of Botulinum Toxin-Containing Products. *Risk C: Monitor*

Capreomycin: May increase neuromuscular-blocking effects of Aminoglycosides. *Risk C: Monitor*

CARBOplatin: Aminoglycosides may increase ototoxic effects of CARBOplatin. This is of particular concern with higher doses of carboplatin. CARBOplatin may increase nephrotoxic effects of Aminoglycosides. Management: Consider alternatives to this combination when possible. If combined, monitor closely for increased ototoxicity and nephrotoxicity. *Risk D: Consider Therapy Modification*

Cardiac Glycosides: Gentamicin (Systemic) may increase serum concentrations of Cardiac Glycosides. *Risk C: Monitor*

Cephalosporins: May increase nephrotoxic effects of Aminoglycosides. Cephalosporins may decrease serum concentrations of Aminoglycosides. *Risk C: Monitor*

Cholera Vaccine: Antibiotics may decrease therapeutic effects of Cholera Vaccine.

Management: Avoid cholera vaccine in patients receiving systemic antibiotics, and within 14 days following the use of oral or parenteral antibiotics. *Risk X: Avoid*

CISplatin: May increase nephrotoxic effects of Aminoglycosides. CISplatin may increase neurotoxic effects of Aminoglycosides. *Risk X: Avoid*

Colistimethate: Aminoglycosides may increase nephrotoxic effects of Colistimethate.

Aminoglycosides may increase neuromuscular-blocking effects of Colistimethate.

Management: Avoid coadministration of colistimethate and aminoglycosides whenever possible due to the risk of nephrotoxicity and neuromuscular blockade. If coadministration cannot be avoided, monitor renal and neuromuscular function. *Risk D: Consider Therapy Modification*

Cyclizine: May increase ototoxic effects of Aminoglycosides. *Risk C: Monitor*

CycloSPORINE (Systemic): Aminoglycosides may increase nephrotoxic effects of CycloSPORINE (Systemic). *Risk C: Monitor*

Distigmine: Aminoglycosides may decrease therapeutic effects of Distigmine. *Risk C: Monitor*

Fecal Microbiota (Live) (Oral): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Oral). *Risk X: Avoid*

Fecal Microbiota (Live) (Rectal): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Rectal). *Risk X: Avoid*

Foscarnet: May increase nephrotoxic effects of Aminoglycosides. Management: Avoid concomitant use of foscarnet and aminoglycoside antibiotics, unless the expected benefits of combined treatment outweigh the risks. If combined, monitor renal function closely. *Risk D: Consider Therapy Modification*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Antibiotics may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Lactobacillus and Estriol: Antibiotics may decrease therapeutic effects of Lactobacillus and Estriol. *Risk C: Monitor*

Loop Diuretics: May increase adverse/toxic effects of Aminoglycosides. Specifically, nephrotoxicity and ototoxicity. *Risk C: Monitor*

Mannitol (Systemic): May increase nephrotoxic effects of Aminoglycosides. Management: Consider avoiding coadministration of aminoglycosides and mannitol. If combined, consider monitor for increased aminoglycoside toxicity (eg, nephrotoxicity, ototoxicity). *Risk D: Consider Therapy Modification*

Mecamylamine: Aminoglycosides may increase neuromuscular-blocking effects of Mecamylamine. *Risk X: Avoid*

Methoxyflurane: Aminoglycosides may increase nephrotoxic effects of Methoxyflurane. *Risk X: Avoid*

Mycophenolate: Antibiotics may decrease active metabolite exposure of Mycophenolate. Specifically, concentrations of mycophenolic acid (MPA) may be reduced. *Risk C: Monitor*

Netilmicin (Ophthalmic): Aminoglycosides may increase nephrotoxic effects of Netilmicin (Ophthalmic). *Risk X: Avoid*

Neuromuscular-Blocking Agents: Aminoglycosides may increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease excretion of Aminoglycosides. Data only in premature infants. *Risk C: Monitor*

Oxatomide: May increase ototoxic effects of Aminoglycosides. *Risk C: Monitor*

Penicillins: May decrease serum concentrations of Aminoglycosides. Primarily associated with extended spectrum penicillins, and patients with renal dysfunction. *Risk C: Monitor*

Polymyxin B: May increase nephrotoxic effects of Aminoglycosides. Polymyxin B may increase neurotoxic effects of Aminoglycosides. *Risk X: Avoid*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Aminoglycosides. *Risk C: Monitor*

Sodium Picosulfate: Antibiotics may decrease therapeutic effects of Sodium Picosulfate. Management: Consider using an alternative product for bowel cleansing prior to a colonoscopy in patients who have recently used or are concurrently using an antibiotic. *Risk D: Consider Therapy Modification*

Tacrolimus (Systemic): Aminoglycosides may increase nephrotoxic effects of Tacrolimus (Systemic). *Risk C: Monitor*

Tenofovir Products: Aminoglycosides may increase serum concentrations of Tenofovir Products. Tenofovir Products may increase serum concentrations of Aminoglycosides.

Risk C: Monitor

Typhoid Vaccine: Antibiotics may decrease therapeutic effects of Typhoid Vaccine. Only the live attenuated Ty21a strain is affected. Management: Avoid use of live attenuated typhoid vaccine (Ty21a) in patients being treated with systemic antibacterial agents. Postpone vaccination until 3 days after cessation of antibiotics and avoid starting antibiotics within 3 days of last vaccine dose. *Risk D: Consider Therapy Modification*

Vancomycin: May increase nephrotoxic effects of Aminoglycosides. Vancomycin may increase neurotoxic effects of Aminoglycosides. Management: Consider avoiding coadministration of aminoglycosides and vancomycin unless clinically indicated. If coadministered, monitor closely for signs of nephrotoxicity and neurotoxicity. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Gentamicin crosses the placenta and can be detected in the cord blood, amniotic fluid, and neonatal serum (Creatsas 1980; Daubenfeld 1974; Regev 2000).

Outcome data following maternal use of gentamicin during pregnancy have been summarized (Nahum 2006; Yu 2020). Aminoglycosides may cause fetal harm if administered to a pregnant patient. There are several reports of total irreversible bilateral congenital deafness in children whose mothers received another aminoglycoside (streptomycin) during pregnancy. Although serious side effects to the fetus/infant have not been reported following maternal use of all aminoglycosides, a potential for harm exists.

Due to pregnancy-induced physiologic changes, some pharmacokinetic parameters of gentamicin may be altered (Lazebnik 1985; Popović 2007).

Gentamicin is part of a recommended regimen for the treatment of pyelonephritis in pregnant patients. Untreated urinary tract infections during pregnancy are associated with adverse pregnancy outcomes including preterm birth and delivery of low-birth-weight infants. Additional maternal complications associated with pyelonephritis are acute respiratory distress syndrome, anemia, disseminated intravascular coagulation, and sepsis. Treatment with a targeted antibiotic is recommended (ACOG 2023).

Gentamicin is part of a recommended regimen for the intrapartum treatment of intraamniotic infections (chorioamnionitis). Untreated intraamniotic infections may be associated with significant neonatal and maternal morbidity, including long-term complications. Treatment is

provided prior to delivery; one additional dose is indicated following cesarean delivery and, in some cases, following vaginal delivery (ACOG 2017; Jung 2024). Gentamicin may also be appropriate as part of a recommended regimen for the treatment of postpartum endometritis (Mackeen 2015).

Gentamicin is used in the management of plague (*Yersinia pestis*). Untreated infections in pregnant patients may result in hemorrhage (including postpartum hemorrhage), maternal and fetal death, preterm birth, and stillbirth. Limited data suggest maternal-fetal transmission of *Y. pestis* can occur if not treated. Pregnant patients should be treated for *Y. pestis*; parenteral antibiotics are preferred for initial treatment when otherwise appropriate. Gentamicin (in combination with a fluoroquinolone) is recommended for treating pregnant patients with bubonic, pharyngeal, pneumonic, or septicemic plague (CDC [Nelson 2021]).

Breastfeeding Considerations

Gentamicin is present in breast milk (Celiloglu 1994).

- Data related to the presence of gentamicin in breast milk are available from 10 healthy mothers given gentamicin 80 mg IM 3 times/day for 5 days postpartum following cesarean delivery. Breast milk samples were obtained after 4 days of treatment. Mean breast milk concentrations of gentamicin ranged from 0.42 mcg/mL (1 hour); 0.48 mcg/mL (3 hours); 0.49 mcg/mL (5 hours); and 0.41 mcg/mL (7 hours) after the dose. Infants were fed 1 hour after the maternal dose was administered and serum levels were drawn 1 hour after the feeding. Detectable gentamicin levels were found in 5 infant samples with a mean level of 0.41 mcg/mL (range: 0.34 to 0.49 mcg/mL). Gentamicin serum concentrations were below the limit of detection in the remaining five infants (<0.27 mcg/mL). The authors estimated the oral infant dose to be ~85 mcg/kg/day using the average milk concentration (Celiloglu 1994).
- Using the highest milk concentration observed in the study, 0.78 mcg/mL (obtained from one patient 1 hour after the dose), the estimated exposure to the breastfeeding infant would be 0.117 mg/kg/day (relative infant dose 1.56% based on a therapeutic infant dose of 7.5 mg/kg/day).
- In general, breastfeeding is considered acceptable when the relative infant dose of a medication is <10% (Anderson 2016; Ito 2000).

In general, modification of bowel flora may occur with any antibiotic exposure (Chung 2002).

The World Health Organization (WHO) considers gentamicin to be compatible with breastfeeding. Infants should be monitored for thrush and diarrhea (WHO 2002).

The risk for transmission of plague (*Yersinia pestis*) via breast milk is considered low. Patients with pneumonic plague can breastfeed if both the mother and infant are receiving antibiotic treatment or the infant is receiving postexposure prophylaxis, considering the risk of exposure to the drug via breast milk. If the infant is not being treated, breast milk should be expressed for at least 48 hours of maternal antibiotic therapy to limit person-to-person contact with the infant. The expressed breast milk may be given to the infant. Once maternal clinical improvement is observed, direct breastfeeding may resume (CDC [Nelson 2021]).

Monitoring Parameters

Urinalysis, urine output, BUN, serum creatinine, plasma gentamicin levels (as appropriate to dosing method). Test hearing before, during, and after treatment, particularly in those at risk for ototoxicity or who will be receiving prolonged therapy (>2 weeks).

Some penicillin derivatives may accelerate the degradation of aminoglycosides in vitro. This may be clinically significant for certain penicillin (ticarcillin, piperacillin, carbenicillin) and aminoglycoside (gentamicin, tobramycin) combination therapy in patients with significant renal impairment. Close monitoring of aminoglycoside levels is warranted.

Reference Range

Conventional/traditional dosing:

Timing of serum samples: Draw peak 30 minutes after 30-minute infusion has been completed or 1 hour following initiation of infusion or IM injection; draw trough within 30 minutes prior to next dose. Obtain drug levels after the third dose unless renal dysfunction/toxicity is suspected.

Therapeutic levels:

Peak:

Sepsis, pneumonia, other serious infections (including life-threatening infections): 7 to 10 mcg/mL (SI: 14.6 to 20.9 micromole/L) (Drew 2025; Matzke 1983; Zaske 1986).

Urinary tract infections, including pyelonephritis: 4 to 6 mcg/mL (SI: 8.4 to 12.5 micromole/L) (Matzke 1983; Noone 1974); some experts suggest a peak of 2 to 4 mcg/mL (SI: 4.2 to 8.4 micromole/L) is sufficient for uncomplicated urinary tract infection (Drew 2025).

Synergy against gram-positive organisms: 3 to 4 mcg/mL (SI: 6.3 to 8.4 micromole/L) (AHA [Baddour 2015]).

Trough:

Gram-negative infections: <2 mcg/mL (SI: <4.2 micromole/L) (ideal target <1 mcg/mL [SI: <2.1 micromole/L]) (Bertino 1994; Drew 2025; Matzke 1983).

Synergy against gram-positive organisms: <1 mcg/mL (SI: <2.1 micromole/L) (AHA [Baddour 2015]).

Obtain drug levels after the third dose unless renal dysfunction/toxicity suspected.

High-dose extended interval dosing (once-daily dosing): Obtain a random gentamicin level between 6 and 14 hours after the start of the gentamicin infusion. Refer to institution-specific nomogram/policies to determine appropriate dosing interval (Bailey 1997; Nicolau 1995). Alternatively, obtain 2 random gentamicin levels with the first level ≥ 1 to 2 hours after the end of the infusion and the second level 6 to 12 hours later (≥ 1 to 2 half-lives apart); adjust dose and interval to achieve an extrapolated peak concentration of ~15 to 20 mcg/mL (SI: 31.4 to 41.8 micromole/L) and trough concentration ≤ 0.5 mcg/mL (SI: <1 micromole/L) (Buijk 2002; Leggett 2015; McNamara 2001; Mueller 2009; Nicolau 1995; Pagkalis 2011; Prescott 2010). When therapy is continued for 5 days or more, monitor the gentamicin levels once or twice weekly (Bailey 1997; Nicolau 1995).

Intraventricular therapeutic drug monitoring: Limited data available (IDSA [Tunkel 2004]; IDSA [Tunkel 2017]): Prior to administration of the next intraventricular dose, withdraw a sample of CSF. This trough CSF concentration divided by the gentamicin MIC of the isolated bacterial pathogen (inhibitory quotient) should exceed 10 to 20.

Mechanism of Action

Interferes with bacterial protein synthesis by binding to 30S ribosomal subunit resulting in a defective bacterial cell membrane

Pharmacokinetics (Adult Data Unless Noted)

Absorption: IM: Rapid and complete; Oral: Poorly absorbed (<1%) (MacDougall 2011).

Distribution: Primarily into extracellular fluid (highly hydrophilic); high concentration in the renal cortex; minimal penetration to CSF and ocular tissues via IV route.

V_d : Varies with age; increased by edema, ascites, fluid overload; decreased with dehydration:

Neonates: 0.45 ± 0.1 L/kg.

Infants: 0.4 ± 0.1 L/kg.

Children: 0.35 ± 0.15 L/kg.

Adolescents: 0.3 ± 0.1 L/kg.

Adults: 0.2 to 0.3 L/kg (Leggett 2015; Xuan 2004).

CSF:blood level ratio: Normal meninges: $<10\%$; Inflamed meninges: $\leq 25\%$ (MacDougall 2011)

Lung:

Bronchial secretion $C_{\max}$ (peak): serum $C_{\max}$ (peak) ratio: $\sim 50\%$, varies with time and route (IM vs IV) (Pennington 1975).

Epithelial lining fluid $C_{\max}$ (peak): serum $C_{\max}$ (peak) ratio: $\sim 32\%$, varies with time (Heffernan 2019; Panidis 2005; Rodvold 2011).

Protein binding: $<30\%$.

Half-life elimination:

Neonates: <1 week: 3 to 11.5 hours; 1 week to 1 month: 3 to 6 hours.

Infants: 4 ± 1 hour.

Children: 2 ± 1 hour.

Adolescents: 1.5 ± 1 hour.

Adults: ~ 2 hours (Regamey 1973); Renal failure: mean: 41 ± 24 hours; Range: 6 to 127 hours (Dager 2006).

Time to peak, serum: IM: 30 to 90 minutes; IV: 30 minutes after 30-minute infusion (MacDougall 2011); **Note:** Distribution is prolonged after larger doses (≥ 90 minutes after 30- to 60-minute infusion of 7 mg/kg) (Demczar 1997; McNamara 2001; Wallace 2002).

Excretion: Urine ($\geq 70\%$ as unchanged drug).

Clearance: Directly related to renal function.

Neonates: 0.045 ± 0.01 L/hour/kg.

Infants: 0.1 ± 0.05 L/hour/kg.

Children: 0.1 ± 0.03 L/hour/kg.

Adolescents: 0.09 ± 0.03 L/hour/kg.

Adults: 0.06 L/hour/kg (Xuan 2004).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Clearance is decreased in renal impairment.

Anti-infective considerations:

Parameters associated with efficacy: Gram-negative bacilli: Concentration-dependent; associated with $C_{\max}$ (peak)/minimum inhibitory concentration (MIC), goal: ≥ 8 to 10 (IDSA [Tamma 2024]; Kashuba 1999; Moore 1987; Zelenitsky 2003) or AUC_{24}/MIC , goal: 30 to 50 (mild/moderate infection, immunocompetent patient) or 80 to 100 (severe or resistant infection) (Bland 2018; Craig 2011; Drusano 2007; IDSA [Tamma 2024]; Nielsen 2011).

Expected drug exposure in patients with normal renal function:

$C_{\max}$ (peak), postdistributional:

Infants, children, and adolescents (varies with age; values are generalized):

Neutropenic fever: 7 mg/kg: ~17 mg/L (Shankar 1999).

Critically ill: 7 to 8 mg/kg: 15 to 22 mg/L (Lopez 2010).

1.5 to 2.5 mg/kg: ~5 to 6 mg/L (Alsultan 2019; Knoderer 2003).

Adults:

7 mg/kg: ~22 mg/L (Finnell 1998; Xuan 2004).

2 mg/kg: ~6 mg/L (Demczar 1997).

AUC_{24} :

Children and adolescents:

Neutropenic fever: 7 to 8 mg/kg: ~45 to 80 mg•hour/L (Bialkowski 2016).

Adults:

7 mg/kg: ~70 to 110 mg•hour/L (Barclay 1995; Craig 2011; Finnell 1998).

Parameters associated with toxicity: Nephrotoxicity is associated with more frequent administration and elevated $C_{\min}$ (trough) concentrations leading to renal accumulation (Bertino 1993; Rybak 1999; Verpooten 1989); AUC has also been found to be predictive (Rybak 1999).

Postantibiotic effect: Bacterial killing continues after gentamicin concentration drops below the MIC of targeted pathogen; generally ~0.5 to 12 hours, though the actual time of postantibiotic effect varies significantly based on multiple factors including

organism, gentamicin $C_{\max}$ (peak), and concomitant antimicrobial therapy (Craig 2011; Gudmundsson 1993; Lacy 1998; Urban 1997).

Patient Counseling Points

(For additional information see "[Gentamicin \(systemic\): Patient drug information](#)")

What is this drug used for?

- It is used to treat bacterial infections.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Decreased appetite
- Weight loss
- Headache
- Upset stomach or throwing up
- More saliva
- Mouth irritation or mouth sores
- Hair loss
- Joint pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Dizziness or passing out
- Change in balance
- Ringing in the ears, hearing loss, or any other changes in hearing
- Muscle weakness

- Trouble breathing that is new or worse
- A burning, numbness, or tingling feeling that is not normal
- Twitching
- Seizures
- Feeling confused
- Feeling very tired or weak
- Depression
- Change in eyesight
- Fever
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country



For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Garamycin; (AR) Argentina: Gentamicina | Gentamicina richet | Gentamicina Rivero | Gentamina | Glevomicina | Plurisemina; (AT) Austria: Gentamicin B Braun; (AU) Australia: Gentamicin | Gentamicin sulphate | Gentamycin; (BD) Bangladesh: Garamycin | Gentamycin | Gentacin | Gentin | Invigen; (BE) Belgium: Gentamycine b.braun | Gentamycine baxter | Geomycine; (BR) Brazil: Garacin; (CL) Chile: Gentalyn | Gentamicina; (CO) Colombia: Garamicina | Gentabiotic | Gentamicina; (CZ) Czech Republic: Gentamicin braun | Gentamycin k; (DE) Germany: Gentamicin B Braun; (DO) Dominican Republic: Garamicina | Gentamicina | Servigenta; (EC) Ecuador: Gentabrand | Gentamicina | Gentastar; (EE) Estonia: Gentamicin; (EG) Egypt: Garamycin | Regaminol; (ES) Spain: Genta Gobens | Gentamicina Braun | Gentamicina grifol | Gentamicina normon | Genticina | Gevramycin; (FI) Finland: Garamycin | Gentamicin B Braun; (FR) France: Gentamycine b. braun | Gentamycine panpharma; (GB) United Kingdom: Gentamicin | Trav gentamicin; (GR) Greece: Gentamedin; (HK) Hong Kong: Gentamicin; (HR) Croatia: Gentamicin b.braun; (ID) Indonesia: Gentamicin sulfat; (IE) Ireland: Cidomycin | Gentacin; (IN) India: Gentamycin; (IQ) Iraq: Gentamicin; (IT) Italy: Gentalyn | Gentamen | Gentamicina b. braun; (JO) Jordan: Cidomycin | Garamycin; (JP) Japan: Geminimycin | Gentacin | Gentamic.sulf.show; (KR) Korea, Republic of: Gentamicin; (KW) Kuwait: Garamycin; (LB) Lebanon: Garamycin; (LT) Lithuania: Gentamycin k; (LU) Luxembourg: Gentamicin; (LV) Latvia: Gentamycin | Gentamycin k; (MA) Morocco: Gentalline | Gentamen | Gentamycine llorente | Gentosyl | Polybac; (MX) Mexico: Barmicil | Fustermicina | Garalen | Gentacin | Gentamicina delta | Tondex; (MY) Malaysia: Garamycin | Gentamicin | Gentamycin; (NL) Netherlands: Gentamicinum; (NO) Norway: Gensumycin | Gentamicin | Gentamicin braun | Gentamicina; (NZ) New Zealand: Gentamicin; (PA) Panama: Vijomycin; (PE) Peru: Azupel; (PH) Philippines: Adelanin | Gentaxin | Obogen | Rocygen; (PK) Pakistan: Genacyn | Gentamicin | Gentamycin | Refobacin; (PL) Poland: Gentamicin b.braun | Gentamycin; (PR) Puerto Rico: Garamycin | Gentamicin; (PT) Portugal: Garalone | Gentamicina | Gentamicina b. braun; (PY) Paraguay: Gentamedin; (QA) Qatar: Genta | Gentam | Gentamed | Genthaver | Genticyn; (RO) Romania: Gentamycin k; (RU) Russian Federation: Gentamicin ferein | Gentamicin k; (SA) Saudi Arabia: Cidomycin | Genta; (SG) Singapore: Gentamicin | Gentamicin sulphate | Gentamycin | Gentacin | Miramycin; (SI) Slovenia: Gentamicin B Braun; (SK) Slovakia: Gentamicim B.braun | Gentamycin k; (SV) El Salvador: Vijomycin; (TH) Thailand: Cidomycin | Garamycin | Gentacin | Gentamicin | Gentamicin injection meiji | Gentamycin | Gentawin | Getamic; (TN) Tunisia: Gentamycine | Gentosyl; (TR) Turkey: Getamisin; (TW) Taiwan: Cantamycin | Garamycin | Genta Gobens | Gentamicin | Gentamycin | Gentamycin sulfate | Gentapromes | Larkmycin | Miramycin | Patamicin | Servigenta; (UA) Ukraine: Gentamycin k; (UY) Uruguay: Gentamina | Migenta; (VE) Venezuela, Bolivarian Republic of: Catogen | Gentalyn injectabl | Gentamicina | Gentisul; (ZA) South Africa: Gentamycin | Gentamycin-fresenius | Sabax gentamix

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